

A Technical Introduction to AlphaFold

Section 1

Derived databases AlphaFill adds cofactors to AlphaFold models where appropriate. This is achieved by searching the Protein Data Bank for similar structures and transplanting cofactors to analogous positions. It is also linked to by UniProt. TmAlphaFold docks AlphaFold models to biological membranes, similar to what OPM does for PDB structures. AFTM uses AlphaFold models to identify transmembrane regions in human proteins, similar to what PDBTM does for PDB structures. ChannelsDB 2.0 uses PDB or AlphaFill models to calculate the pathway a molecule may take to reach an enzyme's active site or to reach another side of a transmembrane pore. The AFDB is not updated with UniProt sequence changes. AlphaSync keeps the AFDB in sync with UniProt entry changes, generating updated structures, residue-level features and contacts. It tries to use an AFDB entry for the exact updated sequence when available and run AlphaFold 2 independently otherwise. It fills in AFDB's blank for large (> 2,700 aa) proteins and proteins with special FASTA characters such as B, Z, U or X. The Encyclopedia of Domains (TED) applies the domain-recognition method from CATH database to 188 million unique structures from the AFDB, identifying nearly 365 million domains, which is 100 million more than what sequence-based methods could identify. The Evolutionary Classification of Protein Domains database (ECOD) assigns ECOD classifications to all SwissProt proteins in AFDB.

Section 2

AlphaFold 1 (2018) was built on work developed by various teams in the 2010s, work that looked at the large databases of related protein sequences now available from many different organisms (most without known 3D structures), to try to find changes at different residues (peptides) that appeared to be correlated, even though the residues were not consecutive in the main chain. Such correlations suggest that the residues may be close to each other physically, even though not close in the sequence, allowing a contact map to be estimated. Building on recent work prior to 2018, AlphaFold 1 extended this by estimating a probability distribution for the distances between residues, effectively transforming the contact map into a distance map. It also used more advanced learning methods than previously to develop the inference. The code was not made publicly available, except to run on sequences of proteins in the 2018 CASP competition.

Section 3

AlphaFold 1, 2, and AlphaFold DB AlphaFold DB provides models of individual protein chains (monomers), rather than their biologically relevant complexes. Many protein

regions are predicted with low confidence score, including the intrinsically disordered protein regions. AlphaFold-2 was validated for predicting effects of point mutations on structure and free energy, with a partial success.

Section 4

CASP14 In November 2020, DeepMind's new version, AlphaFold 2, won CASP14. Overall, AlphaFold 2 made the best prediction for 88 out of the 97 targets. On the competition's preferred global distance test (GDT) measure of accuracy, the program achieved a median score of 92.4 (out of 100), meaning that more than half of its predictions were scored at better than 92.4% for having their atoms in more-or-less the right place, a level of accuracy reported to be comparable to experimental techniques like X-ray crystallography. In 2018 AlphaFold 1 had only reached this level of accuracy in two of all of its predictions. 88% of predictions in the 2020 competition had a GDT_TS score of more than 80. On the group of targets classed as the most difficult, AlphaFold 2 achieved a median score of 87. Measured by the root-mean-square deviation (RMS-D) of the placement of the alpha-carbon atoms of the protein backbone chain, which tends to be dominated by the performance of the worst-fitted outliers, 88% of AlphaFold 2's predictions had an RMS deviation of less than 4 Å for the set of overlapped C-alpha atoms. 76% of predictions achieved better than 3 Å, and 46% had a C-alpha atom RMS accuracy better than 2 Å, with a median RMS deviation in its predictions of 2.1 Å for a set of overlapped CA atoms. AlphaFold 2 also achieved an accuracy in modelling surface side chains described as "really really extraordinary". To further validate AlphaFold 2, the conference organizers approached four leading experimental groups working on structures they found particularly challenging and had been unable to determine. In all four cases the three-dimensional models produced by AlphaFold 2 were sufficiently accurate to determine structures of these proteins by molecular replacement. These included target T1100 (Af1503), a small membrane protein studied by experimentalists for ten years. Of the three structures that AlphaFold 2 had the least success in predicting, one was an unusual multidomain complex consisting of 52 identical copies of the same domain. For all targets with a single domain, excluding only one very large protein and the two structures determined by NMR, AlphaFold 2 achieved a GDT_TS score of over 80.

Section 5

AlphaFold 3 Across several benchmarks, AlphaFold3 has demonstrated, on average, superior performance to conventional search-based docking algorithms in predicting small-molecule-protein binding modes. AlphaFold 3 version can predict structures of protein complexes with a very limited set of selected cofactors and co- and post-translational modifications. Between 50% and 70% of the structures of the human proteome are incomplete

without covalently-attached glycans. Studies have shown that although AlphaFold3 can jointly model protein–ligand co-folding, its accuracy drops markedly on test cases with low similarity to its training data—an area of particular importance for drug discovery. Other work has found that AlphaFold is insensitive to adversarial decoys generated by altering the physicochemical properties of binding pockets, suggesting potential reliance on training-set memorization rather than genuine chemical awareness.

Section 6

Source code Open access to source code of several AlphaFold versions (excluding AlphaFold 3) has been provided by DeepMind in 2022 after requests from the scientific community. The source code and weights of AlphaFold 3 were made available for non-commercial use to the scientific community upon request in November 2024. It became publicly available in February 2025, still retaining the non-commercial restriction.

Section 7

Announced on 8 May 2024, AlphaFold 3 was co-developed by Google DeepMind and Isomorphic Labs, both subsidiaries of Alphabet. AlphaFold 3 is not limited to proteins, as it can also predict the structures of protein complexes with DNA, RNA, post-translational modifications and selected ligands and ions. AlphaFold 3 introduces the "Pairformer," a deep learning architecture inspired by the transformer, which is considered similar to, but simpler than, the Evoformer used in AlphaFold 2. The Pairformer module's initial predictions are refined by a diffusion model. This model begins with a cloud of atoms and iteratively refines their positions, guided by the Pairformer's output, to generate a 3D representation of the molecular structure. The AlphaFold server was created to provide free access to AlphaFold 3 for non-commercial research. As of November 2025, the AlphaFold 3 research paper has been directly cited more than 9,000 times.